# Oliver Johnson

owj4@case.edu | (203) 851–3068 | LinkedIn

## Education

**Case Western Reserve University** — *Cleveland, OH*
*B.S. Physics, Biophysics Concentration | Expected May 2029* — *August 2025–Present*

- GPA: 4.00/4.00; 2026 SSRP Awardee; Dean's High Honors (2×); Varsity Swimming & Diving

**The Hotchkiss School** — *Lakeville, CT*
*High School Diploma* — *August 2021–May 2025*

- High Honor Roll (4×); Dormitory Proctor; Varsity Swimming Captain; Senior Faculty Prize

## Research Experience

**Wang Research Group, Case Western Reserve University** — *Cleveland, OH*
*Undergraduate Research Assistant* — *October 2025–Present*

- Develop **DEMON** (Dynamic Exploration of Molecular Simulations Through Optimized Navigation), an RL-guided trajectory-splitting framework for rare-event exploration that learns directly from molecular trajectories without a hand-designed collective variable.
- Implemented discrete Soft Actor-Critic algorithm and developed training infrastructure including checkpointing, warm-start, replay buffer management, and evaluation pipelines.
- Implemented and benchmarked molecular trajectory encoders combining E(3)-equivariant PaiNN/EGNN spatial representations, rotary temporal Transformers, and permutation-equivariant cross-replica attention.
- Reimplemented the model-system Langevin simulation pipeline in JAX using a JIT-compiled lax.scan integrator and vmap-parallel replica stepping, improving simulation throughput by up to $\approx 190\times$ versus the prior implementation.
- Extended OpenMM parallel-walker execution and implemented graph-aware, memory-mapped replay supporting fixed and variable molecular topologies.
- Diagnosed replica-order dependence by reordering walkers and comparing predicted Q-values, then redesigned the trajectory encoder so permuting walkers produces the corresponding permutation of action values.

## Selected Presentations

**Johnson, Oliver**; Brahmasandra, Arnav; Kosliya Muhammed, Sulekha; Wang, Yihang. "DEMON: Dynamic Exploration of Molecular Simulations Through Optimized Navigation." Poster presentation, Case Western Reserve University Intersections, July 2026.

## Work Experience

**Mailbox Restoration Service** — *New Canaan, CT*
*Founder & Operator* — *Summer 2025*

- Founded and operated an on-demand mailbox-restoration service following identification of the underserved market for express restorations; set pricing models to account for uncertain market size, scheduled client work, and ran targeted digital and print marketing campaigns.

**New Canaan Land Trust** — *New Canaan, CT*
*Summer Stewardship Intern* — *Summer 2024*

- Maintained a 400-acre trail network; designed trail lighting for special events and helped revise parking locations to optimize land use.

## Honors & Awards

2026 CWRU Sponsored Summer Research Program Awardee | 2025 Senior Faculty Prize, The Hotchkiss School
Five-time Connecticut Swimming Top 16 Athlete | 2025 Connecticut Swimming Scholar Athlete | Three-time YMCA National Championship Qualifier

## Technical Skills & Experience

**ML & RL:** Python, PyTorch, JAX, NumPy, discrete SAC
**Computing:** Linux, Slurm, HPC, GPU/MPS workflows
**Simulation:** OpenMM, molecular dynamics
**Engineering:** Performance profiling, Git/GitHub